**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

## MARK SCHEME for the May/June 2013 series

# 4024 MATHEMATICS (SYLLABUS D)

**4024/21** Paper 2, maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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**Abbreviations**

cao correct answer only
cso correct solution only
dep dependent
ft follow through after error
isw ignore subsequent working
oe or equivalent
SC Special Case
www without wrong working
soi seen or implied

## SECTION A

| Qu. | Answers | Mark | Part Marks |
|---|---|---|---|
| **1 (a)** | $x = 3$ | 2 | **M1** for $\pm 5x = \pm 15$ = |
| **(b)** | $x = 4, y = -1$ | 3 | **B2** for one correct value www |
| **(c) (i)** | –1, 0, 1 | 1 | |
| **(ii)** | $y > -2$ final answer | 2 | **B1** for –2 seen |
| **2 (a)** | 24 | 2 | **B1** for 15 seen |
| **(b) (i) (a)** | $180 - q$ cao | 1 | |
| **(i) (b)** | $p - q$ cao | 1 | |
| **(ii) (a)** | 8 cm | 1 | |
| **(ii) (b)** | 4.9 cm | 1 | |
| **3 (a) (i)** | 10, 12 | 1 | |
| **(ii)** | $2m$ oe | 1 | |
| **(b) (i)** | 25, 36 | 1 | |
| **(ii)** | $n^2$ | 1 | |
| **(iii)** | 18 | 1 | |
| **(c) (i)** | $t^2 + 2t$ oe | 1 | |
| **(ii)** | 675 | 1 | |

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| | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | $\frac{10}{11}, \frac{1}{11}, \frac{9}{10}, \frac{1}{10}, \frac{10}{10}, \frac{0}{10}$ oe correctly placed | 2 | **B1** for 3 correct values correctly placed |
| | (ii) (a) | $\frac{6}{11}$ oe | 1 | |
| | (ii) (b) | $\frac{9}{22}$ | 2 | **M1** for 3 × |
| | (b) (i) | 1 | 1 | |
| | (ii) | 2 | 1 | |
| 5 | (a) (i) | € 216 | 1 | |
| | (ii) | ($1 = €) 0.68 | 1 | |
| | (b) (i) | Profit $43.3(0) | 3 | **B2** for Loss $43.40 or<br>**M1** for two of 87.50, 48.60 and $27.20 and<br>**M1** for attempt at adding any three prices and then subtracting 120 |
| | (ii) | 36 to 36.1% | 1 ft | |
| 6 | (a) (i) | 68.7° | 2 | **M1** for $\tan A = \frac{18}{7}$ |
| | (ii) | 257 to 257.5 | 4 | **M1** for $\tan 55 = \frac{18}{DE}$<br>**A1** for $DE$ = 12.6 to 12.61 cm<br>**M1** for $\frac{1}{2}(9 + 7 + \textit{their}\,12.6) \times 18$ or for a complete alternative method |
| | (b) | 26° | 2 | **M1** for 41.5 or 112.5 used |
| 7 | (a) | 0.01 m/s cao | 2 | **M1** for 200/19.94 or 100/9.98 |
| | (b) (i) | $\frac{120}{x}$ or $\frac{120}{x+3}$ | 3 | **B1** |
| | | $\frac{120}{x} - \frac{120}{x+3} = \frac{6}{60}$ oe<br>Correct eqn with denominator removed | | **B1** |
| | (ii) | $x$ = 58.5 or – 61.5 | 3 | **B2** for 1 correct answer<br>Or for 58 – 59 AND –61 – –62<br>**B1** for $\frac{-3 \pm \sqrt{14409}}{2}$<br>**C1** for –58.5 AND 61.5 |
| | (iii) | 123 – 123.1 minutes | 2 | **M1** for 120/*their* positive 58.5 |

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**SECTION B**

| | | | |
|---|---|---|---|
| **8** **(a)** **(i)** | $-5.5$ or $-5\frac{1}{2}$ | 1 | |
| **(ii)** | $f^{-1}(x) = \frac{2x+3}{4}$ | 2 | **C1** for $\frac{2x-3}{4}$ or $\frac{2y+3}{4}$ oe |
| **(iii)** | $g = 0.5$ or $\frac{1}{2}$ | 2 | **M1** for $\frac{8g-3}{2} = g$ |
| **(b)** **(i)** | Enlargement<br>Scale factor –3, Centre $A$ | 2 | **B1**<br>**B1** |
| **(ii)** | 2.2 to 2.24 or $\sqrt{5}$ | 1 | |
| **(iii)** | $\begin{pmatrix} 0 \\ -7 \end{pmatrix}$ | 2 | **B1** for 0<br>**B1** for –7 |
| **(iv)** | $\begin{pmatrix} 10 \\ 1 \end{pmatrix}$ | 2 | **M1** for use of $\overrightarrow{DF} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$ or $\overrightarrow{AF} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ |
| **9** **(a)** **(i)** | $h = 29.8$ to $29.85$ | 2 | **M1** for $\pi \times 4^2 \times h (= 1500)$ |
| **(ii)** | 100 | 1 | |
| **(b)** | $x = 2.5$ | 2 | **M1** for $\frac{1}{2} \times 12x \times 5x$ or better for cross section |
| **(c)** **(i)** | $(2y - 3)(2y + 11)$ | 1 | |
| **(ii)** | $y = 1.5$ or $-5.5$ | 1 | |
| **(iii)** | 67.5 cm$^2$ | 1 ft | |
| **(iv)** | 495 cm$^2$ | 3 | **B1** 2 × *their* **(iii)**<br>**B1** 240 × *their* 1.5 |
| **(d)** | $\frac{9}{25}$ cao | 1 | |

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|---|---|---|---|
| **10 (a) (i)** | $\frac{2}{3}, \frac{2}{3}$ oe | 1 | |
| **(ii)** | 8 points correctly plotted and one set of 5 joined with a curve | 2 ft | **B1** for at least 6 correct plots |
| **(iii)** | 1.7 to 1.8 AND –1.7 to –1.8 | 1 ft | **M1** for tangent to curve at –1.5 soi<br>After M0, SC1 for 3 to 4 |
| **(iv)** | –2.5 to –5 (dep on M1) | 2 | **M1** for $x + y = 2$ drawn |
| **(v)** | –1.3 to –1.4 (dep on M1) | 2 ft | One mark for each |
| **(b) (i)** | $a = 3$, $b = 405$ (cao) | 2 | |
| **(ii)** | (0, 5) cao | 1 | |
| **(iii)** | 20 | 1 ft | |
| **11 (a) (i)** | 510 – 520 m | 1 | |
| **(ii)** | $C$ positioned 7 cm from $A$ and 6 cm from $B$ with both construction arcs drawn | 2 | **B1** for $c$ positioned 7 cm from $A$ and/or 6 cm from $B$ |
| **(iii)** | 146° ±2 | 1 ft | |
| **(iv)** | $D$ positioned 10.3 cm ±0.8 from $A$ and $D\hat{A}C = 34°$ ±2° | 2 | **B1** for DAC = 34 (±2°) |
| **(b) (i)** | 164 to 164.11° www | 4 | **B3** for $QPR$ = 110 to 110.11<br>Or<br>**B2** for $\frac{-2750}{80000}$ or –0.343 to 0.344<br>Or<br>**B1** for (cos $P$ =) $\frac{160^2 + 250^2 - 340^2}{(2\times)160\times 250}$<br>And<br>**M1** for their $P$ + 54° |
| **(ii)** | 18780 – 18800 | 2ft | **M1** for $\frac{1}{2}\times 250\times 160\times \sin 110.1$ |

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| | | | |
|---|---|---|---|
| **12 (a) (i)** | 14.8 kg www | 3 | **M1** for 15×3+14×8+20×12+24×15+31×17+24×20+12×26 (= 2076)<br>**M1** for dividing by 140 (indep) |
| **(ii)** | Correct histogram | 3 | **M2** for 5 correct bars<br>or **M1** for 3 correct bars or all correct heights seen |
| **(iii)** | $\frac{11}{35}$ oe | 2 | **M1** for 15 + 14 + 15 ( = 44) used |
| **(b) (i)** | 9 | 1 | |
| **(ii)** | 35% | 1 | |
| **(iii)** | 96° | 2 | **M1** for (15 + 2) ÷ 64 (× 360) |

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